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Soil Microbiome Contextual Data Specification: Interim Needs Assessment Report

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1 WHY A SOIL MICROBIOME SPECIFICATION

Soil microbiome studies generate rich contextual data: where and how a sample was taken, what had happened at the site, and how the sample was stored, extracted and sequenced. This information is rarely recorded in a consistent, machine-readable form, so sequence data from different projects cannot be reliably compared, combined or reused, and the effort of integrating them falls on each new user.

CIDGOH is developing a harmonised contextual data specification for soil microbiome samples under the Climate Smart Data Collaboration Centre (CS-DCC). This needs assessment asks three questions: what contextual data the literature reports, how well existing standards such as the MIxS soil extension capture it, and what data producers collect and need in practice. This interim report summarises the findings so far and how they will shape the specification. Findings are preliminary; whether each field is required, recommended or optional will be decided once interviews are complete.

2 HOW WE ASSESSED NEEDS

Literature review. A PubMed search for terrestrial soil microbiome sequencing studies reporting contextual data (last 5 years) was narrowed from 5,815 results to 76 primary research abstracts, which were coded against 11 research themes; 17 full-text papers have been read in detail so far. Amplicon-only studies are in scope, as all four research groups interviewed use amplicon sequencing.

Standards review. The MIxS soil extension and the NCBI MIMARKS soil package were reviewed field by field, alongside related schemas, soil information systems and ontologies.

Interviews. Semi-structured interviews (60–90 minutes, 11 topics from project scope to future needs) were held between July and October 2026 with five data producers: three academic research groups, one federal government research team and one industry group carrying out contaminated-site assessment. Two interviews were transcribed from handwritten notes and one awaits a follow-up session.

3 WHAT WE FOUND

3.1 Literature

Of the 76 abstracts, 67 (88%) characterised microbial community composition and 50 addressed nutrient cycling (Figure 1). Most carry several themes, so the specification must serve agricultural, natural, contaminated and experimental soils.

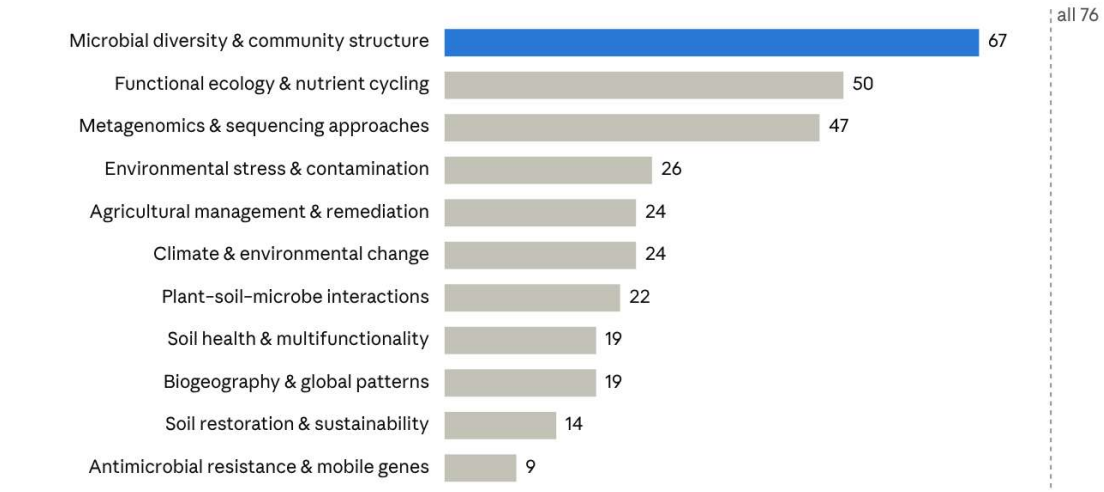


Figure 1. Abstracts tagged with each research theme (n = 76; most carry several themes).

Full-text review found contextual data in six recurring groups: site and climate; soil physico-chemistry; biological measures beyond sequencing, such as enzyme activity and qPCR gene abundance; management and treatments; sampling and processing; and sequencing methods. Four gaps recur:

- **Methods are detached from values.** pH, phosphorus and other properties are measured by different methods, but the method and units are rarely kept with the value.
- **History is narrative.** Land use, management and disturbances such as fire or flooding are described in prose, often without dates.

- **Analysis outputs appear as context.** Diversity indices and multifunctionality scores are reported alongside measured soil properties.
- **Data are hard to reuse.** Supplementary tables often lack data dictionaries.

3.2 Existing standards

The MIxS soil extension from the Genomic Standards Consortium is the most widely used standard and the basis of the NCBI MIMARKS soil package that repositories require. It has 58 soil fields, only four of them required. Both share five structural issues (Table 1) that make data hard to validate, map between systems or interpret years later. Key soil properties (P, K, mineral N, salinity, cation exchange capacity, bulk density) have no dedicated field, and there are no fields for sample type (bulk vs rhizosphere), compositing or extraction kit.

Table 1. Structural issues shared by MIxS soil and MIMARKS soil.

Issue	Examples	Consequence
Several values packed into one field	*crop_rotation*, *agrochem_addition*, *heavy_metals* and *previous_land_use* combine values, units and dates in one string	Custom parsing is needed before data can be validated or mapped
Free text where structure is needed	Temperature, soil carbon, vegetation and storage conditions accept free text; method fields take only a PMID, DOI or URL	Values, units and terms vary across datasets
Relative time	“Current” and “previous” land use and vegetation change meaning as data age	Site history cannot be reconstructed reliably
Uneven controlled vocabularies	ENVO terms are recommended for environment fields, but management and history fields are descriptive text	Hard to tell when two datasets mean the same thing
Events recorded as attributes	Fire and flooding are a date only, with no duration, severity or source	Management and disturbance histories cannot be compared

Other resources fill parts of these gaps, but none covers soil science, management history, sample lifecycle and sequencing together (Table 2).

Table 2. Other resources the specification can draw on.

Resource	What it offers the specification	Limitation
NMDC submission schema	MIxS soil in LinkML, with DataHarmonizer templates and worked examples	US-centric; ontology alignment in progress
NEON soil microbial protocols	Standardised field sampling aligned to GSC and ENVO	Observatory-specific; carbon and biodiversity focus
CS-DCC project schemas	Fields Canadian projects already use	Not harmonised with each other; mostly free text
Ontario Soil Information System (OSIS)	Canadian soil classification, horizons, texture, drainage and slope	No microbiome or sequencing context
MIEM guidelines	Reporting of primers, controls and bioinformatics	Generic across environments

Resource	What it offers the specification	Limitation
GRDI-AMR One Health specification	Ontology-driven sample lifecycle model and DataHarmonizer template	Not soil-specific
Ontologies (ENVO, ChEBI, UO, PATO, GLOSIS)	Terms for environments, chemicals, units and observations	Competing OWL and SKOS approaches to reconcile

3.3 Interviews

Three of the five interviewees record contextual data in spreadsheets with mostly free-text values, and none applies a metadata standard when data are collected; standards are applied, if at all, at repository submission. Table 3 summarises the interviews by topic, with interviewees identified by sector.

Domain	INT-01 Research: agricultural soils (Jul 2026)	INT-02 Research: native grassland (Jul 2026)	INT-03 Industry: contamination site assessment (Sep 2026)	INT-04 Government: farm pathogen and AMR (Sep 2026)	INT-05 Research: organic amendments (Oct 2026)
Objective	Microbial inoculants to cut fertiliser use; community analysis of agricultural soils	Ecosystem services of native grassland plants; effects of conversion to grazing	Contamination assessment for property transactions, insurance and clean-up (Phase 1–3)	Pathogen and AMR transmission across farm matrices; testing mitigation and interventions	Effect of organic amendments (compost, insect-farm frass) on soils
Sample design	Opportunistic in longer-term work; biological duplicates only	Standardised plan: ~20 plants × ~54 sites (~2,000 samples); bulk soil control every 5 samples	Phase 1 history check, then boreholes and monitoring wells; plan adapts to findings	Formal plan powered to the hypothesis; 4 fields, 3 sites; 5 samples pooled per 3 m plot; ≥2 years	Not discussed
Sample type and depth	Homogenised 10 cm cores; bulk vs rhizosphere not separated	45 cm core split 0–15 / 15–30 / 30–45 cm; rhizosphere-associated soil, litter, plant, flowers, pan traps	Continuous soil cores (4–5 ft samplers), groundwater, soil vapour; depth set on site	Top 10 cm bulk soil, forest litter; also manure, water, feed, crops, animals, abattoir	Soil and amendment material; some collected by growers (scooping)
Identifiers	Sequential IDs (GC1, GC2…); no subsample IDs	Concatenated plant–site–plot–subsample ID; LIMS planned	Free-text borehole and depth names	Sample ID linking tables in an SQL database	Group, experiment, sample and sequencing IDs
Location	Lat/long estimated by phone; withheld publicly for landowner privacy	Site, row and lat/long; locations sensitive	GPS waypoint; hand-drawn site map measured from fixed features	Phone GPS (±3 m), stake flags, tagged trees and rocks	Not discussed
Pre-sampling history and events	Urea and nitrification inhibitor treatments; no wider farm history	Past fires noted; adjacent pasture/agriculture	Phase 1 check of past land use (e.g. gas stations, mechanics, storage tanks); rain noted	Treatments by product, date and dose; rotations; manure; geese intrusion; possible legacy PFAS	Amendment treatments
Soil and environmental measurements	C, N, pH pending from collaborator; no weather data	Organic and inorganic C, N, pH; plant cover and IDs; precipitation and temperature	VOCs, hydrocarbons, trichloroethylene; pH; texture, colour, smell; in-situ probe screening	N, P, K, Cu, Zn, Mg; pH; moisture; national climate data; biomass and enzyme activity planned	Not discussed
Storage and handling	Fridge short-term, sometimes −20 °C; ambient transport; collector and dates recorded	Field cooler/freezer ≤5 days, then −20/−80 °C	Jarred with minimal oxygen and frozen; chain of custody kept	Sieved, then frozen at −20/−80 °C (glycerol for counts); no freeze/thaw	Not discussed
Extraction and QC	Qiagen kits; recorded in lab books, not metadata; Qubit, NanoDrop, gel	Magnetic-bead kit (high-throughput), PowerSoil for Nanopore; standardised across sites	Not applicable	Matrix-specific protocols on protocols.io and GitHub; details at follow-up	Not discussed

Domain	INT-01 Research: agricultural soils (Jul 2026)	INT-02 Research: native grassland (Jul 2026)	INT-03 Industry: contamination site assessment (Sep 2026)	INT-04 Government: farm pathogen and AMR (Sep 2026)	INT-05 Research: organic amendments (Oct 2026)
Assays	16S, ITS, shotgun metagenomics (external Illumina); Nanopore for isolates	16S, ITS, AMF 18S, CO1; Nanopore and PacBio metagenomics	Chemical analysis; core imaging	Metagenomics, amplicon and whole genome sequencing	16S amplicon (in-house PCR; external Illumina)
Bioinformatics	QIIME 2, DADA2, SILVA; MEGAHIT/metaSPAdes, binning, GTDB-Tk, DRAM	nf-core amplicon pipeline; SILVA 138.2, UNITE v10, MaarjAM; custom long-read MAG workflows	Not applicable	To be covered at follow-up	DADA2; SILVA and UNITE with versions tracked
Data capture and sharing	Shared spreadsheet, mostly free text; occasional MIxS conversion; NCBI at publication	Excel with shared project headers, mostly free text; NCBI metagenome package	Split across organisations; some projects confidential (defence, nuclear)	SQL database; shared with other agencies and the public; Indigenous land may restrict publication	Spreadsheet on lab drive; no standard; repository plans uncertain
Who captures the data	Research team; collaborator (soil chemistry); sequencing facility (library prep)	Research team; sequencing centres; colleague (soil classification)	Client/consultant (history, field log, chain of custody); drilling contractor (cores, wells); commercial lab (chemistry); imaging company (core images)	Research team; agronomist (nutrient tests, prescriptions); national climate service; accredited labs (toxicity); collaborators (gas flux)	Research team; growers (some sampling); sequencing provider
Core fields named	Time of year, environmental site, soil type, pH, C/N, storage, extraction method	Database and software versions (mandatory); plant biomass and productivity; real-time climate	Not asked; notes emphasise contaminant results, borehole and depth, site history, site map, chain of custody	Treatment product, date, dose; linked sample IDs; location precision; climate; agronomic profile; storage conditions	Diversity and richness outputs (analysis, not context)

Table 3. Interview results by topic. INT-03 and INT-04 were transcribed from handwritten notes; INT-04 awaits a follow-up from library preparation onward. “Not discussed” marks topics not covered.

Seven themes stand out:

- **Consistency is the main pain point.** Headers and values vary by project; even shared headers carry free text.
- **Sample derivation varies and defines comparability.** Depths (10 cm cores, 0–15 cm splits, 4–5 ft cores), compositing and bulk vs rhizosphere separation differ and are not always recorded.
- **Site history is wanted but rarely structured.** Interviewees most want management history, climate and soil chemistry; the one team that records treatments does so as product, date and dose.
- **Lab and bioinformatics detail sits outside the metadata.** Extraction kits, storage, primers and software or database versions live in lab books or with sequencing providers.
- **Several parties create one sample's record.** Contractors, consultants, labs, agronomists and climate services each hold part of the context, and contaminated-site work adds chain of custody and contaminant results.
- **Location and access can be restricted.** Landowner privacy, confidential defence and nuclear sites, and Indigenous land limit what can be published.
- **Soil sits among other sample types.** Farm studies link soil to manure, water, feed, animal and abattoir samples.

4 HOW THIS WILL INFORM THE SPECIFICATION

The findings map onto the design patterns CIDGOH applies to modular data specifications: each module represents one real-world event; it records who did what to which object, when, where, why and how; values are typed and anchored in ontologies; and identifiers link each event to its inputs and outputs. Table 4 shows how each finding will be addressed.

Table 4. From findings to specification design.

Finding	Design pattern	In the specification
Packed fields and free text (3.1–3.3)	One concept per field; typed values and picklists anchored in ontologies	Separate fields for value, unit, date and method; picklists drawing on ENVO, ChEBI and UO
Methods detached from values	Observation: value + unit + method	Measurements recorded as observations linked to the site, sample or collection event (e.g. pH in water vs $CaCl_2$)
Relative time and narrative history	Event-based framing; events, states and plans kept separate	Land use, amendments, management and disturbances recorded as dated events at the site
Variable depth, compositing and subsampling	Traceability through persistent identifiers	Collection, processing and storage events with depth interval, sample type and parent–child sample IDs

Finding	Design pattern	In the specification
Lab and bioinformatics detail outside the metadata	Contextual integrity (how)	Extraction, sequencing and analysis events with kit, primers, platform, software and database versions
Several parties per sample	Contextual integrity (who) and provenance	An actor (person, organisation or role) on every event; links to external lab, agronomic and climate records
Restricted locations	Granularity balance	Coordinate precision, generalised location and data-sharing restriction fields
MIxS seen as too prescriptive, yet required by repositories	Small core plus optional modules; mapping to existing standards	Required core with optional modules, delivered as a DataHarmonizer template that exports to MIxS and NCBI

4.1 Emerging data model

Figure 2 shows the emerging architecture. A small core set of fields is required for every sample. Site history (land use, soil amendments, field management and disturbance) is recorded as dated events at the sampling site rather than as “current” or “previous” values. The sample lifecycle is a chain of events: collection produces the sample; processing and storage can be repeated on the sample or any subsample; and extraction produces the extract that is sequenced and analysed.

Every event module shares the same core fields (who, what, when, where, why and how) plus a free-text details field, and adds only a few fields specific to its type. Measurements are recorded as observations with a value, unit and method, linked to the site, the sample or the collection event. The model is a working draft for discussion with the workgroup.

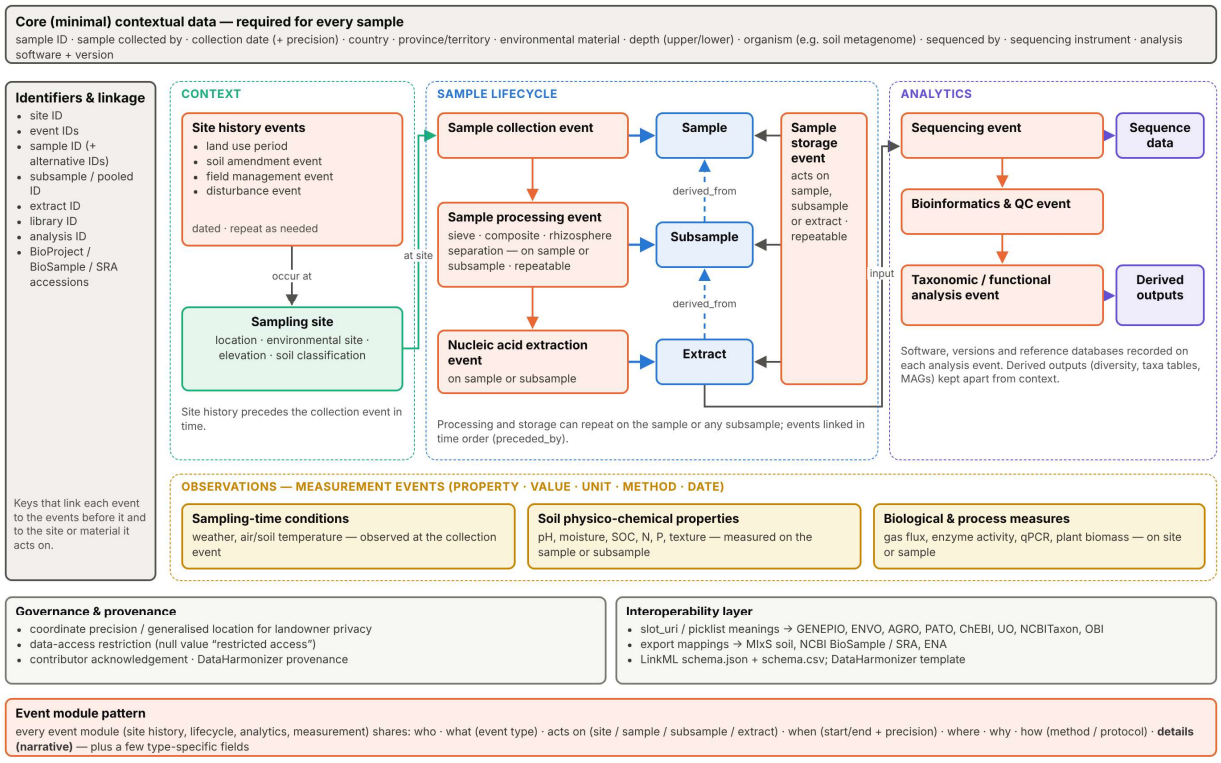


Figure 2. Emerging event-based architecture for the soil microbiome contextual data specification (draft).

5 WHAT HAPPENS NEXT

Further interviews, including soil chemistry and soil classification specialists, will complete the needs assessment, after which each data element will be classed as required, recommended or optional. In parallel, the team will map existing standards field by field, agree an ontology approach with partners, and draft the first modules as a LinkML specification and DataHarmonizer template for testing with participating projects. Feedback on this report and the draft model is welcome.